WASP-36b

R-0736

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_250316` · created 2026-07-28T01:24:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460750.7505 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.048
Transit depth (Rp/Rs)^2	2.3 +/- 1.46 [%]
Orbital Inclination (inc)	80.54 +/- 2.9
Airmass coefficient 1 (a1)	1.27 +/- 0.65
Airmass coefficient 2 (a2)	-0.19 +/- 0.35
Scatter in the residuals of the lightcurve fit is	50.29 %
Best Comparison Star	#1 - [564, 91]
Optimal Aperture	0.75
Optimal Annulus	6.75
Transit Duration (day)	0.045 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.048 vs 0.1368 (deviation 0.32σ)
Duration fit vs published	0.045 d vs 0.0758 d (deviation 0.93σ)
Mid-time O–C	-13.6 min
Depth SNR	3.2
Residual scatter	50.29 %

Light curve

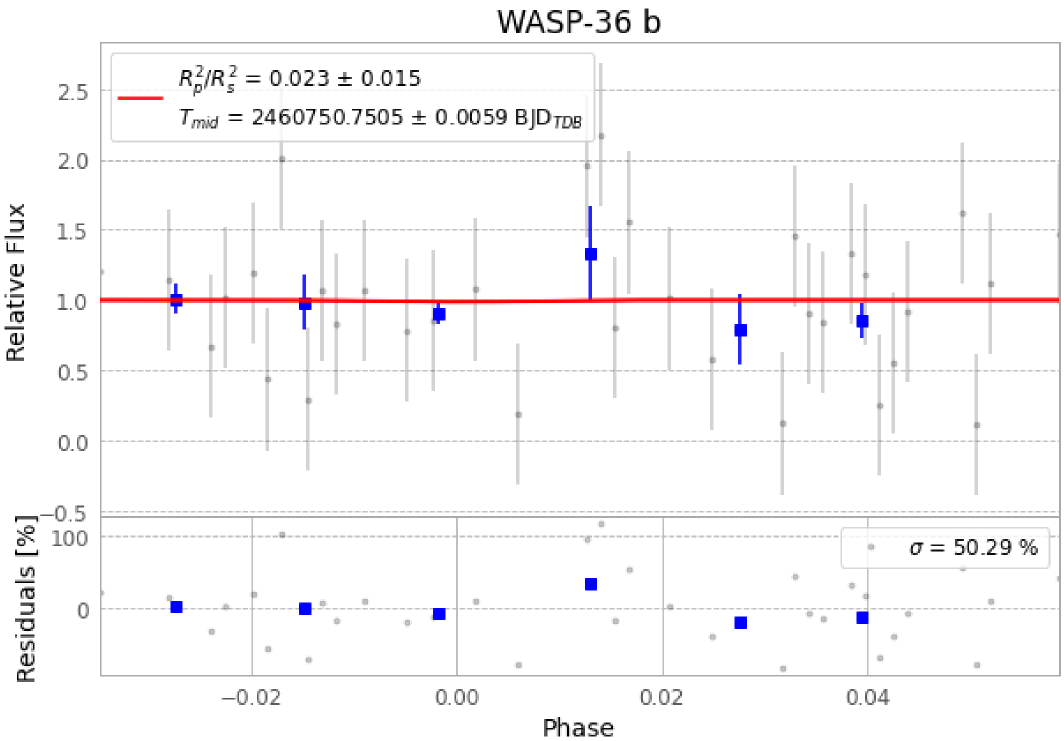

Figure 1 — FinalLightCurve_WASP-36 b_2025-03-15.png (SHA-256 9a57d0941fa652e9…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [536, 446], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2e7a617d21debea2…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

70 FITS frames (46 MB), `WASP-36250316043615.FITS` → `WASP-36250316080315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-250316.log (029c1e3446d6…), FinalLightCurve_WASP-36 b_2025-03-15.png (9a57d0941fa6…), FinalLightCurve_WASP-36 b_2025-03-15.csv (58ea2858c22d…)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-28 01:24Z.